

# Electrical impedance tomography bladder monitoring wearable device

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## ABSTRACT

In this short abstract, we report the recent progress of our bladder monitoring device using electrical impedance tomography. The device's design and image reconstruction algorithms monitor and diagnose lower urinary tract problems. The device is being prepared to the clinical trials, which are planned to start at the end of this year.

## CCS CONCEPTS

• **Hardware** → Printed circuit boards; • **Human-centered computing** → Visualization design and evaluation methods; • **Applied computing** → Life and medical sciences.

## KEYWORDS

electric impedance tomography, medical imaging, wearable device

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## 1 INTRODUCTION

Lower urinary tract conditions are an embarrassing problem for many patients and contribute to a significant deterioration in the quality of life. Therefore, we are developing a wearable device for examining the lower urinary tract to counteract such problems.

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The device is based on two techniques - electromyography (EMG) and electrical impedance tomography (EIT), with which the bladder and parts of the pelvic floor organs can be imaged [1–3].

The device is currently being prepared for the clinical trial phase, whose feedback will allow for improvements and further enhancements. As the trials end, we also intend to develop a smartphone app, which would be capable of registering patient's feedback during the examination. Patients could add any information they find important. During the analysis, by a medical professional, the data from the device would be shown together with collected messages placed in an appropriate place on the timeline. This approach would improve the diagnostics by overviewing the objective data together with patient's feelings.

## 2 HARDWARE

The device consists of textile underwear and an electronic module. The textile part was created in several size variants. The target population of the device to be tested are adult patients, children, and teenagers.

A strip with 16 electrodes for EIT and EMG measurements is attached to the material part using Velcro. A cable connects the electrodes to a small control module that can be placed a convenient place for a patient.

The recorded data logged by the electronic module is downloaded to a computer via a USB interface. The data are processed, and images are reconstructed.

## 3 RECONSTRUCTION

The results obtained with the use of electrical impedance tomography are shown below. We used a Linear (LDA) and Quadratic Discriminant Analysis (QDA) for the bladder reconstruction and compared them to ground truth. The results shown in Figure 2 show the quality of obtained reconstruction (first row). The second row in Figure 2 is very similar to the first one with a small change that the



**Figure 1: Presentation of the device. (left) Complete device with all parts assembled. (right) The device on a dummy.**

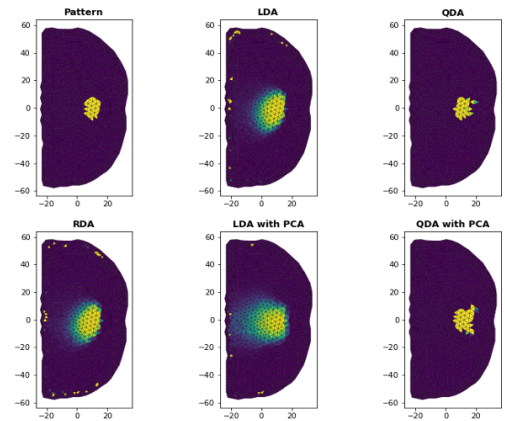
dimension of measurements is reduced with principal component analysis (PCA).

#### 4 PREPARATION FOR CLINICAL TRIALS

The device is currently in preparation for clinical trials. They aim to evaluate the effectiveness of treatment with the device for rehabilitation and evaluation of urinary tract function. The study will involve 2 homogeneous groups of patients: the first group - of women with stress urinary incontinence and the second with urge incontinence. Before qualifying a patient for the research program, each patient will have a gynaecological and urological consultation and diagnostic tests. Ultrasound examination of the urinary tract and reproductive system, general urinalysis, and evaluation of urinary function by Electrical Impedance Tomography. Once the patient is classified into one of the groups, they begin an exercise program with a physiotherapist.

#### 5 CONCLUSION

The developed system is ready for the clinical trials planned for the IV quarter of 2022. During the trial, the patients will wear the device for short (few minutes) and long (few hours) periods. In addition, the device will monitor the patient's bladder and inform them if the level of urine is dropping and the patient is unaware of



**Figure 2: The results obtained from the measurement. First column shows the ground truth, the second and third column shows the results obtained with LDA and QDA, respectively.**

such an incident. The device will be used to monitor and diagnose problems connected to the lower urinary tract, and also to evaluate the progress of the therapy.

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